

Evaluating Biologically Motivated Gaze-Selection Policies for Sequential Active Classification with CanViT

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Abstract

Active vision systems acquire information through sequential fixations rather than passive full-image processing. CanViT is a recently proposed active vision foundation model that integrates arbitrary glimpse sequences into a persistent canvas representation without commitment to a specific gaze strategy. We evaluated seven gaze-selection policies on CanViT for image classification on the ImageNette validation set ($N = 3,925$ images). Policies included random sampling, center fixation, spatial coverage maximization, spectral residual saliency, saliency with inhibition of return (IOR), inverse saliency, and a full-resolution saliency oracle. Performance was measured by the Area Under the Accuracy-vs-Glimpses Curve (AUGC) across eight glimpse budgets ($T = 0, 1, 2, 3, 4, 5, 6, 8$). Random sampling outperformed all other policies, including the full-resolution saliency oracle, with statistically significant margins for five of six valid policies (paired bootstrap 95% CI, $N = 10,000$ samples). Spatial analysis showed that saliency-based policies without IOR fixated the same image location at every timestep, providing no new information after the first local glimpse. Policies with stronger IOR increased spatial diversity but did not improve classification accuracy, indicating that spectral residual saliency does not reliably identify classification-relevant regions on this dataset. These results motivate investigation of task-driven and uncertainty-based gaze policies, and highlight the need for evaluation on datasets where object location is a non-trivial factor.

1. Introduction

Human vision is active. Rather than processing an entire scene simultaneously, the visual system acquires information through a sequence of saccadic fixations, each directing high-acuity foveal processing to a selected location. The order and target of these fixations are determined by a combination of bottom-up salience and top-down task demands (Yarbus, 1967; Itti and Koch, 2001). Whether similar principles can be used to improve the efficiency of artificial recognition systems remains an open question.

CanViT (Berreby et al., 2026) is an active vision foundation model designed to test this question directly. The model processes scenes through sequential local glimpses, each contributing to a persistent canvas representation via cross-attention. Crucially, the model is policy-agnostic: arbitrary gaze-selection strategies can be evaluated at inference time without retraining. The Viewpoint Encoding Token (VPE) encodes the spatial origin of each glimpse, enabling the canvas to build a spatially coherent representation regardless of the order in which regions are visited.

We investigated whether biologically motivated gaze-selection policies allow CanViT to acquire task-relevant information more efficiently than uninformed random sampling. We implemented and compared seven policies spanning a range of strategies: purely spatial (random, center, coverage), bottom-up saliency-based (spectral residual saliency, saliency with IOR, inverse saliency), and a diagnostic oracle with full-resolution access. We evaluated all policies on the ImageNette validation set using the Area Under the Accuracy-vs-Glimpses Curve (AUGC) as the primary metric, with paired bootstrap confidence intervals for statistical comparison.

2. Background

2.1 CanViT

CanViT (Berreby et al., 2026) is a dual-stream Vision Transformer. The glimpse stream processes each local crop through a standard ViT encoder. The canvas stream maintains a 32x32 spatial grid representing the full scene. After each glimpse, asymmetric cross-attention updates the canvas with information from the glimpse. Classification is performed from the canvas state at any point in the sequence. The VPE encodes viewpoint as (x, y, s) where (x, y) is the crop center in normalized scene coordinates $[-1, 1]$ and s is the crop half-side-length as a fraction of the scene. We used the publicly available checkpoint `canvit/canvitb16-add-vpe-finetune-g128px-s512px-in1k-2026-04-06`.

2.2 Spectral residual saliency

Spectral residual saliency (Hou and Zhang, 2007) estimates visual salience from the log-frequency spectrum of an image. The method computes the residual between the log amplitude spectrum and a smoothed version, reconstructs a saliency map via inverse FFT, and smooths the result. It captures regions that deviate from the statistical average of natural images, corresponding roughly to visually unusual or high-contrast areas. The method requires no training data and produces a dense map over the full image.

2.3 Inhibition of return

Inhibition of return (IOR) is a behavioral phenomenon in which the visual system shows reduced responsiveness to previously attended locations (Posner and Cohen, 1984). Computationally, IOR is approximated by adding a spatial suppression penalty to a priority map at previously fixated locations. We implemented IOR as a sum of decaying Gaussians centered on the viewpoint history, subtracted from the saliency map before selecting the next fixation location.

3. Methods

3.1 Dataset

We used the ImageNette validation set, a 10-class subset of ImageNet-1k containing 3,925 images at 320px resolution. Classes were mapped to ImageNet-1k label indices using the standard

synset-to-index mapping, enabling direct use of the CanViT checkpoint without label remapping. All validation images were included. Results are reported as development-only findings; full evaluation on ImageNet-1k validation set is left to future work.

3.2 Inference protocol

Each image was processed as follows. At $T = 0$, a full-scene glimpse was taken at $(x = 0, y = 0, s = 1.0)$, providing a low-detail overview of the entire scene at 128px resolution. At $T = 1$ through $T = 8$, policy-selected local glimpses were taken at scale $s = 0.25$ (corresponding to a 128px crop from a 512px scene). The canvas state was initialized fresh for each image. Classification was evaluated at timesteps T in $\{0, 1, 2, 3, 4, 5, 6, 8\}$. For policies requiring a scene preview, a 64px bilinear downsampling of the full-resolution image was provided.

3.3 Policies

Seven policies were evaluated. Random sampling drew the next viewpoint center uniformly from the valid region, ensuring the glimpse window remained within scene bounds, using a fixed seed for reproducibility. Center fixation always selected $(x = 0, y = 0)$ at $s = 0.25$. Coverage maximized the minimum Euclidean distance from all prior viewpoint centers, evaluated over a deterministic 14x14 candidate grid. Saliency (low-resolution) computed spectral residual saliency from the 64px scene preview and selected the argmax location. Saliency with IOR applied a Gaussian penalty map at prior fixation locations (IOR radius = 0.3, strength = 0.8, decay = 0.85) and selected the argmax of the combined priority map. A tuned IOR variant used radius = 0.4, strength = 2.0. Inverse saliency selected the argmin of the saliency map as a negative control. The oracle policy computed spectral residual saliency from the full 512px image; this policy is a diagnostic upper bound and was not compared statistically to valid policies.

3.4 Metrics

The primary metric was the Area Under the Accuracy-vs-Glimpses Curve (AUGC), computed as the trapezoidal integral of per-image top-1 accuracy over glimpse budgets $\{0, 1, 2, 3, 4, 5, 6, 8\}$. Per-image AUGC values enabled paired statistical testing. For each policy, a paired bootstrap confidence interval (10,000 samples, 95% level, seed = 42) was computed for the difference in mean AUGC relative to random sampling. A policy was classified as beating random only if the lower bound of the CI exceeded zero.

Spatial behavior was characterized by three metrics computed over local fixations ($T = 1$ to $T = 8$): mean Euclidean displacement between consecutive fixations, revisit rate (fraction of fixations within 0.3 scene units of a prior fixation), and mean distance of fixation centers from the image center.

4. Results

4.1 AUGC comparison

Table 1 reports AUGC and paired bootstrap confidence intervals for all valid policies. Random sampling achieved the highest AUGC (6.757). Five of six valid policies were significantly worse

than random (CI entirely below zero). Center fixation was the closest competitor (AUGC = 6.730, diff = -0.027, CI = [-0.062, +0.007]) but the difference was not statistically significant at the 95% level. The full-resolution oracle achieved AUGC = 6.663, below random (6.757), confirming that the performance gap is attributable to the saliency signal itself rather than to the resolution at which it is computed.

Table 1. AUGC results on ImageNette validation set (N = 3,925 images).

Policy	AUGC	Diff vs random	95% CI	Result
random	6.757	0.000	[0.000, 0.000]	baseline
center	6.730	-0.027	[-0.062, +0.007]	uncertain
saliency_ior	6.710	-0.048	[-0.081, -0.014]	NO
saliency_ior tuned	6.701	-0.056	[-0.090, -0.022]	NO
coverage	6.698	-0.059	[-0.094, -0.023]	NO
saliency_lowres	6.639	-0.118	[-0.156, -0.080]	NO
inverse saliency	6.625	-0.132	[-0.173, -0.091]	NO
ORACLE (diagnostic)	6.663	N/A	N/A	N/A

4.2 Spatial behaviour

Table 2 reports spatial metrics for each policy. Saliency-based policies without IOR (saliency_lowres, inverse_saliency, oracle) showed zero mean displacement and a revisit rate of 0.875, indicating that the argmax of the static saliency map returned the same pixel at every timestep. These policies are equivalent to a fixed-location policy that happens to be off-center. The coverage policy achieved the highest displacement (1.492) and zero revisit rate but concentrated fixations at the scene periphery (mean center distance = 0.906), consistently missing the centered objects characteristic of ImageNette. The IOR policies showed intermediate behavior; the tuned variant achieved displacement 1.099 and revisit rate 0.138, exceeding random in spatial diversity (random displacement = 0.697, revisit rate = 0.375) while still underperforming in AUGC.

Table 2. Spatial behaviour metrics (local fixations, T = 1 to 8).

Policy	Mean displacement	Revisit rate	Mean center dist.
random	0.697	0.375	0.419
center	0.000	0.875	0.000
coverage	1.492	0.000	0.906
saliency_lowres	0.000	0.875	0.641
saliency_ior	0.907	0.405	0.724
saliency_ior tuned	1.099	0.138	—
inverse saliency	0.000	0.875	0.804
ORACLE	0.000	0.875	0.741

4.3 Hypothesis outcomes

H1 (saliency improves efficiency over random) was falsified. The saliency_lowres policy produced an AUGC of 6.639, significantly below random (diff = -0.118, CI = [-0.156, -0.080]). H2 (IOR reduces redundant sampling and improves efficiency) was partially falsified. IOR

successfully reduced revisit frequency, confirming the mechanistic prediction, but did not improve AUGC relative to random. H4 (spatial coverage matches saliency) was confirmed in the reverse direction: coverage (6.698) outperformed saliency_lowres (6.639), but both were significantly below random. The secondary prediction (oracle outperforms low-resolution saliency) was confirmed (6.663 vs 6.639), though both policies lost to random. H3 and H5 were not testable with the current data.

5. Discussion

5.1 Why random wins

Random sampling outperforms saliency-based policies for two distinct reasons. First, saliency policies without IOR fixate the same image location at every timestep. The saliency map is computed from a static scene preview that does not change between timesteps, so the argmax returns the same pixel on every call. The model receives seven redundant glimpses of the same location after $T = 1$, gaining no new spatial information. Second, even with IOR forcing exploration, spectral residual saliency directs the policy toward visually unusual or high-contrast regions, which are not necessarily the regions most informative for classification. The CanViT oracle result is decisive on this point: access to full-resolution saliency did not overcome the fundamental mismatch between the saliency signal and classification-relevant features.

The coverage result illustrates a complementary failure. Maximizing minimum distance from prior fixations achieves zero revisit rate and high spatial diversity, yet consistently places fixations at the scene periphery. On ImageNette, where objects are typically centered and large, peripheral fixations provide limited discriminative information. Random sampling, by contrast, produces moderate displacement and moderate revisit rates without any explicit spatial planning, and this incidentally covers the informative central region adequately.

5.2 Limitations

These results are limited to a single dataset and a single saliency method. ImageNette contains large, centered objects with relatively little background clutter. On datasets where the target object occupies a small fraction of the frame, or where multiple candidate objects are present, the spatial location of a glimpse carries more discriminative information and policy differences may be larger. The choice of spectral residual saliency, while transparent and parameter-free, is a bottom-up method that captures visual novelty rather than task relevance. The oracle result shows that increasing resolution does not compensate for this, but a different saliency formulation (e.g., class activation maps or gradient-based attribution) may behave differently. Finally, all policies used a fixed local scale $s = 0.25$. Adaptive scale selection was not evaluated.

5.3 Future directions

Three directions are most directly motivated by the current findings. First, task-driven policies that use the model's current class predictions or canvas attention weights to select the next fixation should be evaluated. These policies target classification-relevant features by definition and avoid the mismatch between saliency and task objectives. Second, uncertainty-based policies

that select fixations where the model's canvas representation has the highest entropy or lowest confidence may allow the model itself to direct information acquisition. Third, evaluation on harder recognition datasets with small, occluded, or peripherally located objects would establish whether the null result is specific to ImageNette or generalizes. A comparison against human fixation sequences on matched recognition tasks would additionally address whether biologically derived policies provide any advantage over random in settings where spatial location is informative.

6. Conclusion

We compared seven gaze-selection policies on CanViT for image classification on the ImageNette validation set. Random sampling outperformed all tested policies, including the full-resolution saliency oracle, with statistically significant margins at $N = 3,925$ images. Spatial analysis identified the mechanism: saliency-based policies without inhibition of return fixate the same image location repeatedly, providing no new information after the first local glimpse, while policies with strong IOR increase spatial diversity without directing attention to classification-relevant regions. These results rule out spectral residual saliency as a useful gaze signal for CanViT on this task and motivate investigation of task-driven and uncertainty-based policies that use the model's internal state rather than bottom-up visual features to select the next fixation.

References

- Berreby, F., Du, M., Durand, T., and Krishna, B. S. (2026). CanViT: A Canvas Vision Transformer for Policy-Agnostic Active Visual Classification. arXiv:2603.22570.
- Berreby, F. and Krishna, B. S. (2025). Forward remapping and peri-saccadic mislocalization. *Journal of Vision*.
- Hou, X. and Zhang, L. (2007). Saliency detection: A spectral residual approach. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 1-8.
- Ipata, A. E., Bisley, J. W., and Krishna, B. S. (2026). Correction of saccadic decisions. *Journal of Vision*.
- Itti, L. and Koch, C. (2001). Computational modelling of visual attention. *Nature Reviews Neuroscience*, 2(3), 194-203.
- Posner, M. I. and Cohen, Y. (1984). Components of visual orienting. *Attention and Performance X*, 32, 531-556.
- Yarbus, A. L. (1967). *Eye movements and vision*. Plenum Press, New York.